Vulnerability Assessment Report

15th September 2026

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 20XX to August 20XX. NIST SP 800-30 Rev. 1 is used to guide the risk analysis of the information system.

Purpose

The database server is a centralized computer system that stores and manages large amounts of data. The server is used to store customer, campaign, and analytic data that can later be analyzed to track performance and personalize marketing efforts. It is critical to secure the system because of its regular use for marketing operations and the severe impact on business operations if data integrity or availability is compromised.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Competitor	*Obtain sensitive information via exfiltration*	*1*	*3*	*3*
Hacker	*Conduct Denial of Service (DoS) attacks*	*3*	*3*	*9*
Employee	*Alter/Delete critical information*	*2*	*3*	*6*

Approach

Risks that were measured considered the data storage and management procedures of the business. Potential threat sources and events were determined using the likelihood of a security incident given the open access permissions of the information system. The severity of potential incidents were weighed against the impact on day-to-day operational needs.

Remediation Strategy

To remediate the identified risks, implement multi-factor authentication, strong password policies, and role-based access control to strictly limit user privileges and prevent unauthorized data alterations. Enable comprehensive database logging and auditing mechanisms to track access, and enforce IP allow-listing to restrict database connections exclusively to corporate office networks while applying rate-limiting to mitigate Denial of Service attacks. Finally, mandate modern TLS encryption for all data in motion, fully deprecating legacy SSL protocols to protect sensitive customer and campaign data from exfiltration.